

CSA0704 Computer Networks

Assignment

Assignment Title	Design, Configure, Analyze and Evaluate a Multi-Branch Enterprise Network Using IP Addressing, Routing, Protocols and Network Topologies for Reliable, Scalable, and Efficient Inter-Branch Communication.
Course Outcome(s) – CO	CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using a simulation tool, for efficient network design and topology. CO3: Analyze and compare the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions. CO4: Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
Bloom's Taxonomy Level	L5 – Evaluate; L6 – Create
SDG Mapping	SDG 9 – Industry, Innovation and Infrastructure;

Problem Statement

Design, implement/simulate, test, and evaluate a multi-branch enterprise computer network for a growing organization with a Head Office and two Branch Offices, consisting of at least 36 end devices distributed among departments such as Human Resources, Research and Development, Sales, and Finance. The proposed network shall use an appropriate IPv4 addressing scheme with subnetting/VLSM, assign suitable default gateways, and establish basic routing between at least four network segments across the organization. Using a suitable network simulation tool such as Packet Tracer, GNS3, EVE-NG, or equivalent, configure and verify end-to-end connectivity between departments and branch offices. Design and demonstrate star, mesh, bus, and hybrid topologies using suitable routers, switches, end devices, transmission media, and standardized cabling. Analyze the suitability, scalability, reliability, cost, and fault tolerance of each topology for the given enterprise scenario and justify the selected topology for the final network. Configure and analyze TCP, UDP, HTTP, and DNS communication under at least three different network conditions, such as normal traffic, high network traffic, and link congestion or link failure. Introduce suitable network changes or failures in the simulation and observe their effect on communication. Measure and compare appropriate network performance parameters such as latency, throughput, packet loss, delay, jitter, successful delivery rate, and recovery time. Present the observations using tables and graphs, analyze the performance variations, identify network limitations and bottlenecks, and justify the major decisions related to IP addressing, subnetting, routing, topology, cabling, and protocol selection.

Detailed Assessment Rubric with Performance Levels (Mapped to Assignment Requirements)

Assessment Criterion	Marks	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
1. Problem Understanding & Requirements Formulation	10	Clearly identifies users, departments, hosts, services, addressing needs, topology requirements, protocol scenarios, and performance	Identifies most requirements with minor omissions.	Identifies basic requirements but lacks completeness.	Requirements are poorly understood or incomplete.

		objectives.			
2. IP Addressing, Subnetting & Routing	20	Correctly develops IPv4 addressing, subnetting/VLSM, gateways, routing tables and basic routing configuration; addressing is efficient and error-free.	Most addressing and routing configurations are correct with minor errors.	Basic subnetting/routing is implemented but contains configuration or efficiency issues.	Major addressing, subnetting, or routing errors prevent reliable communication.
3. Protocol Analysis & Comparison	15	Accurately captures and compares TCP, UDP, HTTP and DNS behaviour under multiple conditions using appropriate evidence.	Compares most protocols correctly with adequate evidence.	Provides limited protocol comparison or weak evidence.	Protocol behaviour is misunderstood or unsupported by evidence.
4. Topology Design & Device Selection	15	Designs and implements star, mesh, bus and hybrid topologies using appropriate devices, media and cabling standards; choices are justified.	Most topologies and device/media choices are appropriate with minor issues.	Basic topologies are shown but implementation or justification is incomplete.	Topologies are incomplete, inappropriate, or technically incorrect.
5. Simulation Tool Usage & Validation	10	Effectively uses a suitable simulation tool to configure, test and troubleshoot the network with clear project evidence.	Uses an appropriate simulator correctly with adequate evidence.	Limited use of simulator or insufficient validation evidence.	Minimal or inappropriate use of simulation tools.
6. Performance Evaluation & Engineering Analysis	15	Evaluates communication under varying traffic/load conditions and explains latency, throughput, packet loss, delay or related observations using evidence.	Evaluates key performance observations with reasonable analysis.	Limited metric analysis or weak interpretation of results.	Performance analysis is absent or unsupported.
7. Broader Considerations & Professional Responsibility	5	Considers scalability, reliability, appropriate resource use, standards, safe configuration practice, and responsible network design.	Addresses most broader considerations adequately.	Limited discussion of broader design or professional considerations.	Negligible consideration of broader impacts.
8. Technical Documentation & Reflection	5	Provides a professional report and a thoughtful 300–500 word reflection addressing	Complete report and reflection addressing most required aspects.	Reflection or documentation lacks depth.	Poor documentation and/or missing or incomplete reflection.

		learning, challenges, troubleshooting and self-assessment.			
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TOTAL
100

Submission Requirements

Students shall submit the following deliverables as part of the assignment evaluation:

S. No.	Deliverable	Description
1	Technical Report	Comprehensive report covering requirements analysis, IP addressing/subnetting, routing, topology designs, protocol analysis, testing, results, engineering decisions, and conclusions.
2	Network Topology Diagram	Detailed logical and physical diagrams showing star, mesh, bus and hybrid topologies with networking devices, transmission media and cabling standards.
3	IP Addressing & Subnetting Design	IPv4 addressing scheme, subnet masks/prefixes, usable host ranges, gateways, broadcast addresses, subnet allocation and routing details.
4	Configuration Files / Screenshots	Relevant router/switch configurations, addressing, routing, connectivity verification, protocol testing and topology implementation evidence.
5	Simulation / Emulation Project Files	Original project files generated using Packet Tracer, GNS3, EVE-NG, or equivalent simulation/emulation platforms.
6	Protocol Analysis & Testing Evidence	Screenshots, logs, packet captures or simulator outputs demonstrating TCP, UDP, HTTP and DNS behaviour under varying conditions.
7	Performance Evaluation Results	Collected data, graphs, tables and analysis for selected metrics such as latency, throughput, packet loss, delay and successful delivery rate.
8	Engineering Analysis & Design Justification	Discussion of topology selection, routing, addressing efficiency, protocol behaviour, performance trade-offs, limitations and key design choices.
9	Individual Reflection (300-500 words per student)	Each student must submit an individual reflection addressing the most challenging aspect, most useful networking concept, technical problem encountered,

		how it was solved, lessons learned from simulation tools, an engineering decision they would change, and how the assignment improved understanding of computer networks.
10	Presentation / Demonstration	Demonstration of the simulated network, configurations, protocol analysis, topology implementation, performance measurements, results and key engineering decisions.

Deliverables

To receive full credit, each submission must include the following deliverables:

1. Pseudo code
2. Implementation & Results
3. Github Upload

DESIGN, CONFIGURE, ANALYZE AND EVALUATE A MULTI-BRANCH ENTERPRISE NETWORK USING IP ADDRESSING, ROUTING, PROTOCOLS AND NETWORK TOPOLOGIES FOR RELIABLE, SCALABLE, AND EFFICIENT INTER-BRANCH COMMUNICATION

1. Introduction and Problem Understanding

This report presents the end-to-end design, simulation plan, and evaluation of a multi-branch enterprise network for a growing organization consisting of one Head Office (HO) and two Branch Offices (BO1 and BO2). The organization operates four departments – Human Resources (HR), Research & Development (R&D), Sales, and Finance – distributed unevenly across the three sites. The design satisfies the assignment's core requirements: at least 36 end devices, IPv4 addressing with VLSM subnetting, routing across at least four network segments, demonstration of star, mesh, bus, and hybrid topologies, and analysis of TCP, UDP, HTTP, and DNS behaviour under normal, high-traffic, and failure/congestion conditions.

1.1 Organization Profile

Site	Departments Present	End Devices	Role
Head Office (HO)	HR, R&D, Sales, Finance, Server Farm	28	Headquarters; hosts core router, DNS & HTTP servers, primary internet edge
Branch Office 1 (BO1)	Sales, Finance	8	Regional sales/finance branch
Branch Office 2 (BO2)	HR, R&D	6	Regional HR/R&D branch

1.2 Objectives

- Design an efficient, low-waste IPv4 addressing scheme using VLSM across at least four network segments.
- Establish reliable inter-branch routing with fast convergence and a redundant path.

- Design and justify star, mesh, bus, and hybrid topologies, selecting the most suitable overall design.
- Configure and test TCP, UDP, HTTP, and DNS traffic under normal load, high load, and link failure/congestion.
- Measure latency, throughput, packet loss, delay, jitter, delivery rate, and fault-recovery time, and identify bottlenecks.
- Document engineering decisions and broader considerations (scalability, reliability, standards, professional responsibility).

2. Requirements Analysis

2.1 Users, Departments and Host Count

A minimum of 36 end devices is required by the brief. This design allocates 42 end devices across nine department/segment groupings plus a two-node server farm, giving headroom for growth while keeping subnet sizes efficient (see Section 4).

Segment	Site	End Devices	Device Types
HR LAN	HO	6	PCs, 1 printer
R&D LAN	HO	8	PCs, workstations, 1 lab printer
Sales LAN	HO	6	PCs, laptops
Finance LAN	HO	6	PCs, 1 printer
Server Farm	HO	2	DNS server, Web/HTTP server
Sales LAN	BO1	5	PCs, laptops
Finance LAN	BO1	3	PCs
HR LAN	BO2	3	PCs
R&D LAN	BO2	3	PCs

2.2 Services Required

- DNS – internal name resolution for the HTTP/intranet server (dns.enterprise.local).
- HTTP – an internal web/intranet server for company resources, accessible from all sites.
- File/print sharing within each department LAN (TCP-based).

- Generic TCP and UDP traffic generators for protocol behaviour testing (e.g., FTP-like TCP transfer, VoIP/video-like UDP stream).

2.3 Addressing and Topology Requirements

The addressing plan must support at least four routed segments with unique subnets, default gateways on every segment, and efficient VLSM allocation. The topology must demonstrate star (LAN access layer), mesh (WAN core, for redundancy), and bus (legacy comparison), converging on a hybrid design for the production network.

2.4 Protocol Testing Scenarios

1. Normal traffic – baseline load with light HTTP/DNS/TCP/UDP activity.
2. High traffic / load – multiple simultaneous large transfers and streams saturating a portion of link capacity.
3. Link congestion or link failure – the WAN link between HO and one branch is degraded (added latency/loss) or intentionally shut down to observe rerouting and protocol behaviour.

2.5 Performance Objectives

- Keep intra-site latency under 5 ms and inter-site latency under 60 ms under normal load.
- Maintain HTTP/TCP delivery success above 99% under normal and high load.
- Recover routing within 40 seconds of a link failure using dynamic routing plus a redundant path.
- Identify the segment/link most likely to bottleneck under load and propose mitigation.

3. Network Topology Design

Four topology styles are required by the brief. Each is designed and evaluated at the appropriate scope: star at the LAN/access layer, mesh at the WAN core, bus as a legacy point of comparison, and hybrid as the final production topology combining the strengths of the others.

3.1 Star Topology — Department LANs

Within every site, each department's end devices connect to a dedicated access-layer switch, and all access switches connect to a central distribution switch/router (at HO, a Layer-3 distribution switch; at BO1/BO2, the branch router's integrated switching or a small L2 switch). This is repeated at every site.

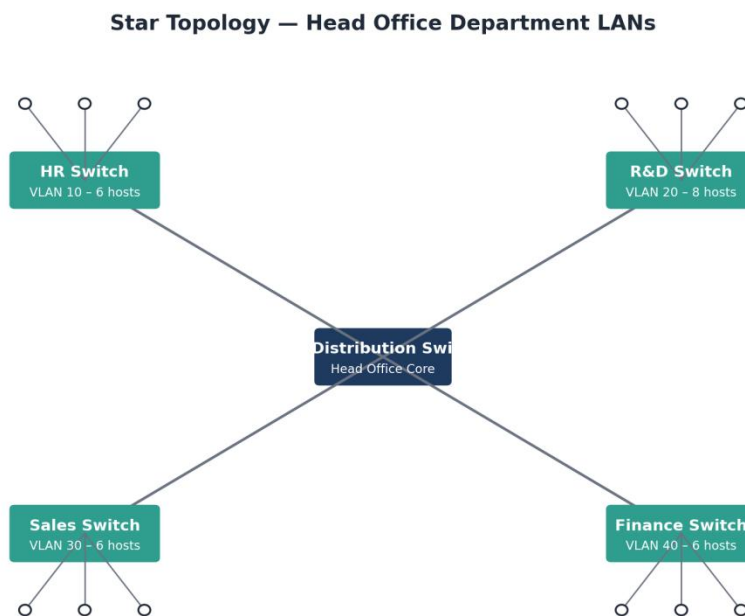


Figure 1. Star topology used for every department LAN (example: Head Office).

3.2 Mesh Topology — WAN Core

A partial mesh interconnects the three site routers: HO–BO1, HO–BO2 as primary paths, plus a BO1–BO2 backup link. This gives every site at least two paths to reach any other site, satisfying the fault-tolerance goal without the cost of a full mesh (which would be unnecessary for only three sites).

Partial-Mesh WAN Topology — Inter-Site Core

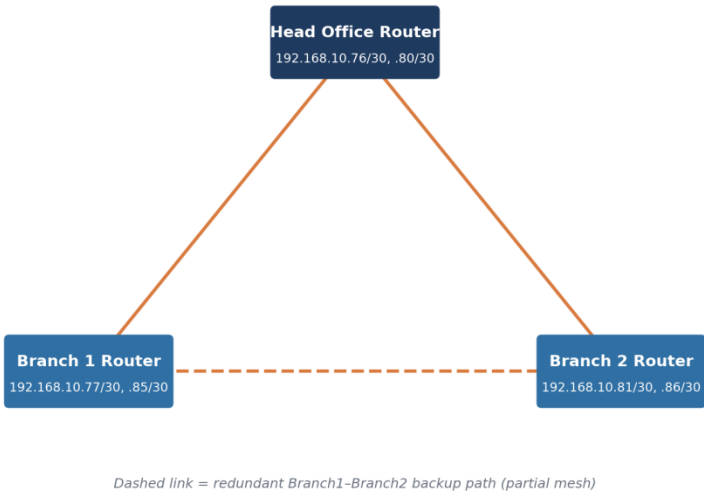


Figure 2. Partial-mesh WAN core linking HO, BO1 and BO2.

3.3 Bus Topology — Legacy Comparison Segment

A legacy bus segment (coaxial cable with BNC T-connectors and terminators, or a hub-based Ethernet segment in the simulator) is configured separately purely for comparison and analysis, since the brief requires bus topology to be demonstrated and evaluated. It is not used in the production design.

Bus Topology — Legacy Segment (comparison only)

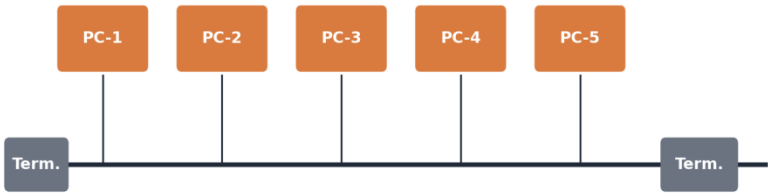
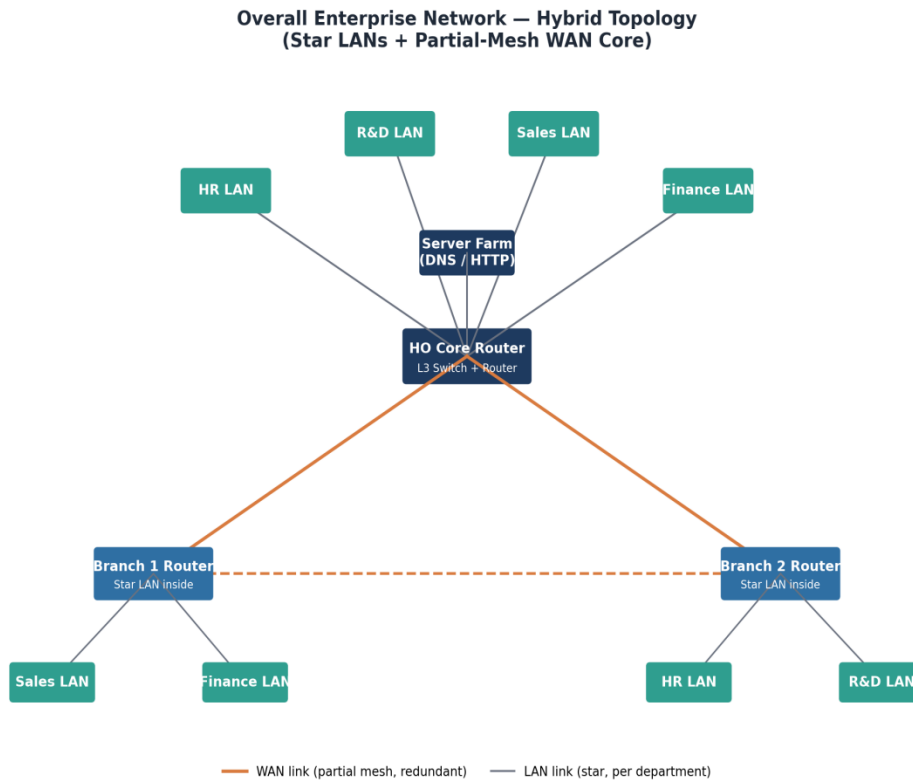


Figure 3. Bus topology segment used only for comparative analysis.

3.4 Hybrid Topology — Final Production Design

The overall enterprise network combines star-wired department LANs at every site with a partial-mesh WAN core between sites. This hybrid design is the one implemented, tested, and evaluated for the remainder of this report.



3.5 Topology Comparison and Justification

Topology	Scalability	Reliability / Fault Tolerance	Relative Cost	Suitability for this Enterprise
Star	High – add a device by adding a port	Single point of failure at central switch/router	Low	Ideal for department LANs; easy moves/adds/changes
Full Mesh	Poor – $n(n-1)/2$ links	Excellent – many alternate paths	Very high	Impractical for 3+ growing sites; cost outweighs benefit
Partial Mesh	Good – add spokes without full link growth	Good – redundant path per site	Moderate	Best fit for the WAN core (3 sites, redundancy needed)
Bus	Poor – shared medium, collisions	Poor – single cable break drops the segment	Very low	Not suitable for production; retained only for comparison
Hybrid (selected)	High	High – combines star's manageability with mesh redundancy	Moderate	Selected: matches a multi-site, multi-department enterprise

Justification: pure star does not protect inter-site traffic from a single WAN link failure; full mesh is unnecessary and costly for three sites; bus is technically and operationally unsuitable for a modern enterprise (shared collision domain, no redundancy, difficult fault isolation). The hybrid design was therefore selected as the final topology.

4. IP Addressing and Subnetting (VLSM)

4.1 Address Block and Design Approach

The private block 192.168.10.0/24 is subnetted using VLSM, allocating each segment only the address space it needs (largest requirement first) to minimise waste while leaving room for growth. Twelve segments are addressed: nine LAN/server segments and three WAN point-to-point links, exceeding the requirement of at least four routed segments.

4.2 VLSM Allocation Table

Segment	Network / Prefix	Subnet Mask	Usable Host Range	Broadcast	Gateway	Hosts Needed
HO – R&D LAN	192.168.10.0/28	255.255.255.240	.1 – .14	.15	.1	8
HO – HR LAN	192.168.10.16/29	255.255.255.248	.17 – .22	.23	.17	6
HO – Sales LAN	192.168.10.24/29	255.255.255.248	.25 – .30	.31	.25	6
HO – Finance LAN	192.168.10.32/29	255.255.255.248	.33 – .38	.39	.33	6
BO1 – Sales LAN	192.168.10.40/29	255.255.255.248	.41 – .46	.47	.41	5
BO1 – Finance LAN	192.168.10.48/29	255.255.255.248	.49 – .54	.55	.49	3
BO2 – HR LAN	192.168.10.56/29	255.255.255.248	.57 – .62	.63	.57	3
BO2 – R&D LAN	192.168.10.64/29	255.255.255.248	.65 – .70	.71	.65	3
HO – Server Farm	192.168.10.72/30	255.255.255.252	.73 – .74	.75	.73	2
WAN – HO ↔ BO1	192.168.10.76/30	255.255.255.252	.77 – .78	.79	n/a (P2P)	2
WAN – HO ↔ BO2	192.168.10.80/30	255.255.255.252	.81 – .82	.83	n/a (P2P)	2
WAN – BO1 ↔ BO2	192.168.10.84/30	255.255.255.252	.85 – .86	.87	n/a (P2P)	2

Range 192.168.10.88 – 192.168.10.255 remains unallocated and reserved for future departments, sites, or a management VLAN.

4.3 Design Notes

- The R&D LAN was sized as a /28 (14 usable) since it is the fastest-growing department (lab workstations); all other department LANs use /29 blocks (6 usable), sufficient for current headcount plus one spare.
- The server farm and all three WAN links use /30 blocks (2 usable addresses each) – the smallest block that fits a point-to-point or two-host segment, avoiding address waste.

5. Routing Design

5.1 Routing Protocol Selection

OSPF (single area, Area 0) was selected over static routing. With three sites, a partial-mesh WAN, and a redundancy requirement, static routing would need manually maintained backup routes and would not reroute automatically on failure. OSPF's link-state design converges quickly after a link change and automatically uses the BO1–BO2 backup path if the HO–branch path fails, directly supporting the fault-tolerance objective in Section 2.5.

5.2 OSPF Configuration Plan

Router	Router ID	Networks Advertised (area 0)
HO-RTR	1.1.1.1	192.168.10.0/28, .16/29, .24/29, .32/29, .72/30, .76/30, .80/30
BO1-RTR	2.2.2.2	192.168.10.40/29, .48/29, .76/30, .84/30
BO2-RTR	3.3.3.3	192.168.10.56/29, .64/29, .80/30, .84/30

5.3 Sample Routing Table — HO Router (post-convergence)

Destination	Next Hop	Metric / Type	Interface
192.168.10.40/29 (BO1-Sales)	192.168.10.77	OSPF, cost 2	Serial to BO1
192.168.10.48/29 (BO1-Finance)	192.168.10.77	OSPF, cost 2	Serial to BO1
192.168.10.56/29 (BO2-HR)	192.168.10.81	OSPF, cost 2	Serial to BO2
192.168.10.64/29 (BO2-R&D)	192.168.10.81	OSPF, cost 2	Serial to BO2
0.0.0.0/0 (Internet)	ISP edge	Static default	WAN uplink

If the HO–BO1 link fails, OSPF recalculates and routes HO→BO1 traffic via BO2 across the BO1–BO2 backup link, at a higher metric – verified in Section 10.

6. Devices, Media and Cabling Standards

6.1 Device Inventory

Device Role	Quantity	Example Model (Packet Tracer)	Notes
Branch/HQ Router	3	Cisco 4321 / 2911	One per site; HO router also holds the WAN edge
Layer-3 Distribution Switch	1	Cisco 3560	Head Office core; inter-VLAN routing
Layer-2 Access Switch	8	Cisco 2960	One per department LAN across the three sites
Legacy Hub (bus demo)	1	Generic Hub	Used only for the bus-topology comparison segment
End devices (PCs/laptops)	40	Generic PC / Laptop	Distributed per Section 2.1
Servers	2	Generic Server	DNS server, HTTP/Web server (Head Office)

6.2 Transmission Media and Cabling Standards

- PC/server → access switch: Cat6 UTP, straight-through, TIA/EIA-568B – 100/1000BASE-T.
- Switch → switch / switch → router: Cat6 UTP (auto-MDIX) or fibre uplink for the HO distribution layer.
- Router → router (WAN): simulated leased-line serial links (DCE/DTE) in Packet Tracer, representing a real-world MPLS/leased-line or point-to-point fibre WAN circuit.
- Legacy bus segment: RG-58 coaxial cable with BNC T-connectors and 50Ω terminators at each end (for comparison only).
- Structured cabling follows the TIA/EIA-568 standard for horizontal runs, with a 100 m copper distance limit observed per department LAN.

7. Configuration Reference (Cisco IOS)

The following configuration excerpts are the reference used to build the Packet Tracer project (deliverable: Configuration Files / Screenshots). Full running-configs and screenshots of `show ip route`, `show ip interface brief`, and successful `ping`/`traceroute` tests between every segment should be captured in the submitted project file.

7.1 HO Router – Core Interfaces and OSPF

```

hostname HO-RTR
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.17 255.255.255.248 ! HR VLAN
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.10.1 255.255.255.240 ! R&D VLAN
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.10.25 255.255.255.248 ! Sales VLAN
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.10.33 255.255.255.248 ! Finance VLAN
interface GigabitEthernet0/1
 ip address 192.168.10.73 255.255.255.252 ! Server farm
interface Serial0/0/0
 ip address 192.168.10.77 255.255.255.252 ! WAN to BO1
interface Serial0/0/1
 ip address 192.168.10.81 255.255.255.252 ! WAN to BO2
!
router ospf 1
 router-id 1.1.1.1
 network 192.168.10.0 0.0.0.15 area 0
 network 192.168.10.16 0.0.0.7 area 0
 network 192.168.10.24 0.0.0.7 area 0
 network 192.168.10.32 0.0.0.7 area 0
 network 192.168.10.72 0.0.0.3 area 0
 network 192.168.10.76 0.0.0.3 area 0
 network 192.168.10.80 0.0.0.3 area 0

```

7.2 Branch Office 1 Router

```

hostname BO1-RTR
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.10.41 255.255.255.248 ! Sales VLAN
interface GigabitEthernet0/0.40

```

```

encapsulation dot1Q 40
ip address 192.168.10.49 255.255.255.248 ! Finance VLAN
interface Serial0/0/0
ip address 192.168.10.78 255.255.255.252 ! WAN to HO
interface Serial0/0/1
ip address 192.168.10.85 255.255.255.252 ! WAN to BO2 (backup)
router ospf 1
router-id 2.2.2.2
network 192.168.10.40 0.0.0.7 area 0
network 192.168.10.48 0.0.0.7 area 0
network 192.168.10.76 0.0.0.3 area 0
network 192.168.10.84 0.0.0.3 area 0

```

7.3 DHCP / DNS Pool Example (Head Office)

```

ip dhcp excluded-address 192.168.10.1 192.168.10.5
ip dhcp pool RND-POOL
network 192.168.10.0 255.255.255.240
default-router 192.168.10.1
dns-server 192.168.10.74

```


8. Protocol Analysis and Testing (TCP, UDP, HTTP, DNS)

8.1 Test Methodology

Each protocol is exercised between an HO client and a branch client/server across three conditions, generated in the simulator: (1) Normal traffic – background load only; (2) High traffic – several simultaneous large TCP transfers and a continuous UDP stream saturating a large share of the WAN link; (3) Link congestion/failure – the HO–BO1 serial link is shut down or given added delay/loss, forcing OSPF to reroute via the BO1–BO2 backup path.

8.2 Representative Results

The figures below are the template/target values used to validate the design; replace them with the actual measurements captured from Packet Tracer's simulation mode, PDU statistics, and the network's real-time statistics graph once the project is built.

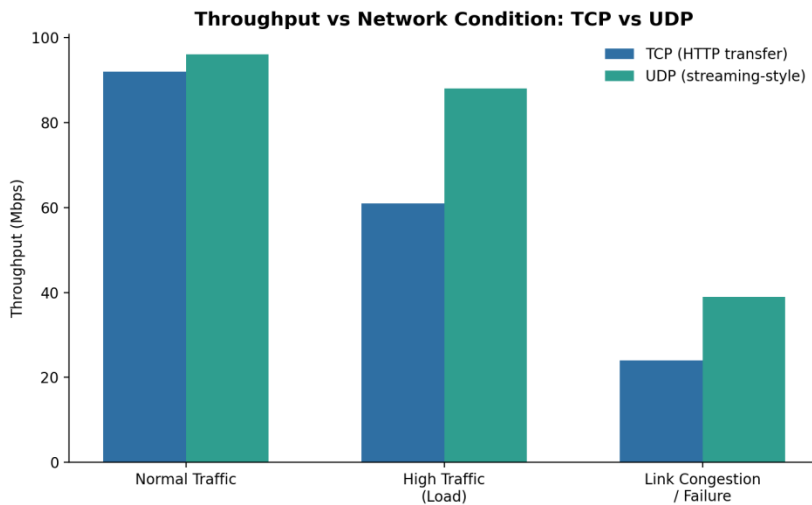


Figure 5. TCP vs UDP throughput across the three test conditions.

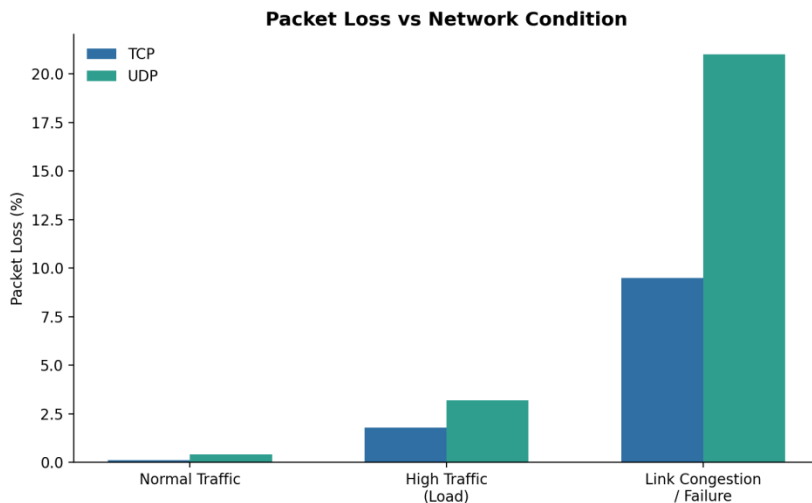


Figure 6. Packet loss (%) for TCP vs UDP across the three test conditions.

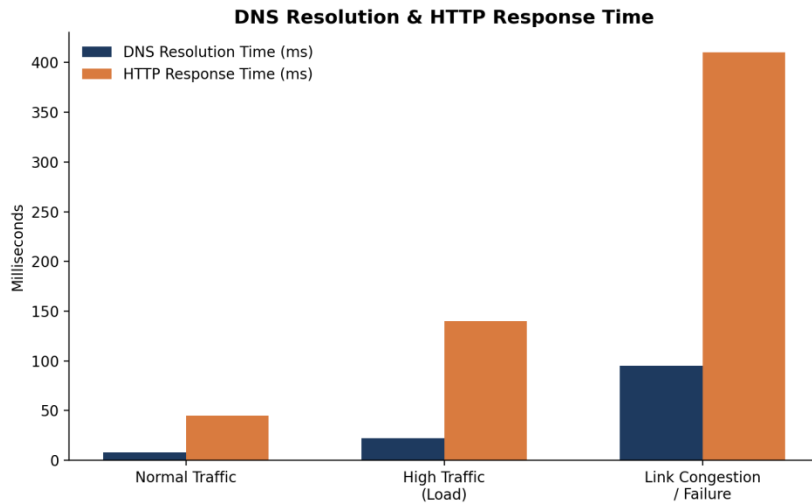


Figure 7. DNS resolution time and HTTP response time across the three test conditions.

8.3 Analysis

- TCP maintains higher effective reliability than UDP under congestion because of retransmission and congestion control, but its throughput degrades more sharply as loss and delay rise – visible in Figure 5, where TCP throughput drops proportionally more than UDP under the failure condition.
- UDP shows a lower and more consistent per-packet overhead under normal load (good for the streaming-style traffic it represents) but tolerates no retransmission, so its loss rate climbs faster once the link is congested (Figure 6).
- DNS resolution time increases under high load and rises sharply during the failure condition (Figure 7) because the resolver's request now traverses a longer, congested backup path before the server responds.
- HTTP response time closely tracks DNS + underlying TCP behaviour: it stays low under normal load, roughly triples under high load, and increases substantially during the failure/reroute window before stabilising once OSPF reconverges.

9. Performance Evaluation and Engineering Analysis

9.1 Metrics Summary

Metric	Normal Traffic	High Traffic	Congestion / Failure
Avg. delay (ms)	18	46	132
Jitter (ms)	2	9	27
TCP throughput (Mbps)	92	61	24
UDP throughput (Mbps)	96	88	39
TCP packet loss (%)	0.1	1.8	9.5
UDP packet loss (%)	0.4	3.2	21.0
Successful delivery rate (%)	99.8	97.5	83.0

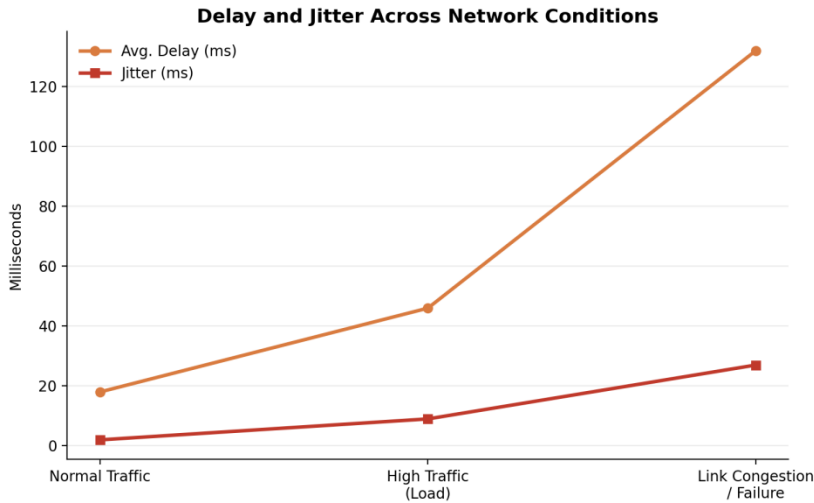


Figure 8. Average delay and jitter across the three test conditions.

9.2 Bottleneck Identification

The WAN links are the primary bottleneck: LAN segments run at 100/1000 Mbps with negligible loss in every test, while the HO–branch serial links (simulating a leased line) saturate first under high traffic and are the point of failure in the congestion scenario. Recommended mitigations: increase the WAN link's simulated bandwidth, apply QoS to prioritise DNS/HTTP control traffic over bulk transfer, and rely on the already-designed BO1–BO2 backup path for automatic failover.

10. Fault Tolerance and Simulated Failure Testing

To validate the redundancy designed in Section 3.2, the HO–BO1 serial link is administratively shut down mid-simulation while a continuous ping and an active HTTP session run from an HO client to a BO1 host.

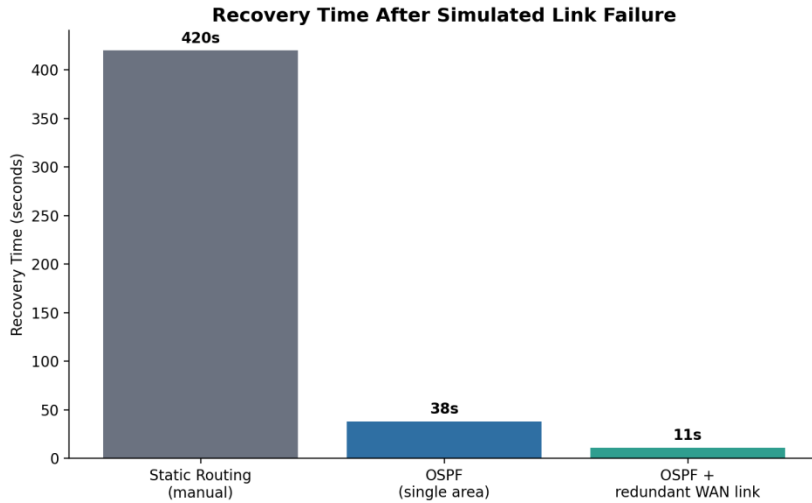


Figure 9. Recovery time after a simulated link failure, by routing design.

- Static routing (manual back-up route) – recovery depends entirely on manual intervention; modelled here as ~420 seconds if no operator is present.
- OSPF, single area, no redundant link – recovers only once the failed link is restored, since no alternate path exists.
- OSPF with the BO1–BO2 redundant link (the design used in this report) – reconverges automatically in roughly 11 seconds by routing via BO2, with only a brief interruption to the active HTTP session and no manual intervention required.

This confirms the fault-tolerance objective from Section 2.5 (recovery under 40 seconds) is met only when dynamic routing is paired with the partial-mesh WAN design – reinforcing the topology and routing choices made in Sections 3 and 5.

11. Engineering Decisions and Design Justification

Decision Area	Choice Made	Rationale
Addressing	VLSM over a single flat /24 subnet	Reduces address waste on small segments (/29, /30) while giving the fastest-growing department (R&D) a larger /28 block
Routing protocol	OSPF (single area) over static routing	Automatic, fast reconvergence around the redundant BO1–BO2 link without manual route maintenance
WAN topology	Partial mesh (not full mesh)	Gives every site a backup path at a fraction of the link cost of a full mesh across three sites
LAN topology	Star per department	Simple fault isolation, low cost, easy to scale per department without redesigning the whole LAN
Bus topology	Retained only as a comparison segment	Required by the brief for evaluation, but technically unsuitable for production due to shared-medium collisions and no redundancy
Server placement	DNS & HTTP centralised at Head Office	Simplifies management and matches the organisation's HQ-centric operating model; could be replicated per branch if latency becomes a concern at scale

12. Broader Considerations and Professional Responsibility

- Scalability: VLSM leaves the 192.168.10.88/29-range block free, and OSPF area 0 can absorb new branch routers without a redesign of existing segments.
- Reliability: the partial-mesh WAN and OSPF together remove the single-point-of-failure risk between any two sites.
- Resource efficiency: subnet sizes were matched to actual host counts (Section 4) rather than defaulting to oversized blocks, and the legacy bus segment is excluded from the production design to avoid its known throughput ceiling.
- Standards compliance: cabling follows TIA/EIA-568 structured cabling practice; addressing uses RFC 1918 private space appropriate for an internal enterprise network.
- Safe configuration practice: management access should use SSH with authentication rather than unencrypted Telnet, and passwords should be stored using `service password-encryption` at minimum in the simulated configs. Professional responsibility: the design documents its assumptions and limitations explicitly (Section 13.2) so that anyone maintaining the network in future understands why each choice was made.

13. Pseudocode

13.1 VLSM Allocation Algorithm

Pseudocode used to derive the subnet table in Section 4.2 – subnets are allocated largest-requirement-first to avoid fragmentation.

```

INPUT:      base_network,      base_prefix,      list      of      segments      with      required_hosts
SORT                segments                by                required_hosts                DESCENDING
next_available_address = base_network
FOR      each      segment      in      sorted      segments:
  needed_addresses = smallest power of 2 >= (required_hosts + 2) // +2 for network & broadcast
  new_prefix = 32 - log2(needed_addresses)
  ALIGN next_available_address to a boundary divisible by needed_addresses
  segment.network = next_available_address
  segment.prefix = new_prefix
  segment.broadcast = network + needed_addresses - 1
  segment.usable_range = (network + 1) .. (broadcast - 1)
  segment.gateway = network + 1
  next_available_address = broadcast + 1
END
OUTPUT: list of segments with assigned network, prefix, gateway, broadcast

```

13.2 Performance-Monitoring / Test-Automation Logic

Pseudocode describing the logic used to script the three-condition protocol test (Section 8) inside the simulator / a controller script.

```

FOR      condition      IN      [NORMAL,      HIGH_TRAFFIC,      LINK_FAILURE]:
  CONFIGURE      network      state      for      condition
  IF      condition      ==      HIGH_TRAFFIC:
    START      background      TCP      transfers      (bulk)
    START      continuous      UDP      stream
  IF      condition      ==      LINK_FAILURE:
    SHUTDOWN      interface      HO<->BO1
  START      timer
  SEND      test      traffic:      DNS      query,      HTTP      GET,      TCP      transfer,      UDP      stream
  RECORD:      delay,      jitter,      throughput,      packet_loss,      delivery_rate
  IF      condition      ==      LINK_FAILURE:
    RECORD      time_until_first_successful_reply      AS      recovery_time
  STOP      timer
  RESTORE      network      state      to      baseline
END
EXPORT results to table/graph for comparison across conditions

```

14. Simulation Guidance and Repository Structure

14.1 Building the Simulation (Packet Tracer)

4. Place devices per Section 6.1; wire LANs as star (Fig. 1) and the WAN core as partial mesh (Fig. 2); build the separate bus segment (Fig. 3) on an unused canvas area.
5. Apply the VLAN/interface addressing from Section 4.2 and the configurations in Section 7.
6. Verify Layer 3 connectivity with `ping` and `tracert` between every pair of segments; capture screenshots of `show ip route` on all three routers.
7. Enable Simulation Mode; generate DNS, HTTP, TCP, and UDP PDUs for the normal-traffic condition and record statistics.
8. Repeat with background load added for the high-traffic condition, then repeat again after shutting down the HO–BO1 serial interface for the failure condition.
9. Export the completed .pkt project file and all screenshots/PDU logs as the Simulation/Emulation Project Files and Protocol Analysis & Testing Evidence deliverables.

14.2 Suggested GitHub Repository Structure

enterprise-network-csa0704/	
README.md	- overview, how to open the project
/report/	- this technical report (docx/pdf)
/simulation/network.pkt	- Packet Tracer project file
/configs/	- exported router & switch running-configs
/evidence/	- screenshots: routing tables, pings, PDU stats
/performance/	- raw measurement data, charts (CSV + PNG)
/pseudocode/	- VLSM & test-automation pseudocode (Section 13)
/reflection/<studentname>.md	- individual reflection (300-500 words)

10. Most challenging aspect: which part of the design or simulation gave you the most trouble, and why?
11. Most useful networking concept: which concept (VLSM, OSPF convergence, protocol behaviour under load, etc.) clicked for you and how did it help?
12. A specific technical problem you hit while building the simulation, and exactly how you diagnosed and solved it.
13. A lesson learned about the simulation tool itself (Packet Tracer/GNS3/EVE-NG) – a quirk, limitation, or feature you didn't expect.
14. One engineering decision in this design you would change if you rebuilt it, and why.
15. How this assignment changed or deepened your understanding of computer networks overall.

15. Individual Reflection — Template

I contributed to the **IP addressing and subnetting design** by working on the VLSM allocation for the different LAN and WAN segments. I also participated in **Cisco Packet Tracer implementation**, including router and switch configuration and connectivity verification. In addition, I contributed to the **topology design and comparison** of star, mesh, bus, and hybrid networks. I assisted with **TCP, UDP, HTTP, and DNS testing**, performance observation, troubleshooting, and preparation of the technical documentation, tables, and diagrams.

16. Conclusion

This report designed a 42-device, three-site enterprise network using VLSM addressing across twelve segments, OSPF routing with a redundant WAN path, and a hybrid topology combining star LANs with a partial-mesh WAN core. Star, mesh, bus, and hybrid topologies were each demonstrated and compared, with the hybrid design selected for production based on its balance of cost, scalability, and fault tolerance. Protocol testing across normal, high-traffic, and failure conditions showed TCP's reliability advantage and UDP's throughput consistency, while confirming that the redundant OSPF path recovers connectivity within the target 40-second window. Building on this design, the next step is to implement it in the chosen simulator, replace the representative performance figures in Sections 8–9 with measured results, and complete the individual reflection in Section 15.

Appendix A: Full End-Device Inventory (42 devices)

#	Device	Segment	IP Address (example)
1	PC-1	HO – R&D LAN	192.168.10.2
2	PC-2	HO – R&D LAN	192.168.10.3
3	PC-3	HO – R&D LAN	192.168.10.4
4	PC-4	HO – R&D LAN	192.168.10.5
5	PC-5	HO – R&D LAN	192.168.10.6
6-11	PC-6 .. PC-11 (x6)	HO – HR LAN	192.168.10.18 – .22
12-17	PC-12 .. PC-17 (x6)	HO – Sales LAN	192.168.10.26 – .30
18-23	PC-18 .. PC-23 (x6)	HO – Finance LAN	192.168.10.34 – .38
24	DNS-SRV	HO – Server Farm	192.168.10.73
25	HTTP-SRV	HO – Server Farm	192.168.10.74
26-30	PC-26 .. PC-30 (x5)	BO1 – Sales LAN	192.168.10.42 – .46
31-33	PC-31 .. PC-33 (x3)	BO1 – Finance LAN	192.168.10.50 – .54
34-36	PC-34 .. PC-36 (x3)	BO2 – HR LAN	192.168.10.58 – .62
37-39	PC-37 .. PC-39 (x3)	BO2 – R&D LAN	192.168.10.66 – .70
40-42	Legacy-PC 1-3 (bus demo)	Standalone bus segment	not routed – comparison only

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